Informatica®

Informatica® Cloud Data Integration

SAP HANA Connector

Informatica Cloud Data Integration SAP HANA Connector
April 2026

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Preface

Use *SAP HANA Connector* to learn how to read from or write data to SAP HANA database by using Cloud Data Integration. Learn to create an SAP HANA connection, develop mappings, and run mapping tasks and data transfer tasks in Cloud Data Integration.

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https://network.informatica.com/community/informatica-network/products/cloud-integration

Developers can learn more and share tips at the Cloud Developer community:

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CHAPTER 1

Introduction to SAP HANA Connector

You can use SAP HANA Connector to read from or write data to SAP HANA databases deployed in the cloud or on-premises. SAP HANA Connector uses the JDBC driver at design time and the ODBC driver at run time.

You can use SAP HANA Connector to perform the following operations:

- Read or lookup data from SAP HANA database tables and views.
- Read data from external views that are created by SAP BW CompositeProviders in the SAP system. For more information about how to create views for CompositeProviders, see Generating SAP HANA Views for CompositeProviders in the SAP documentation.
- Read data from the following HANA database modelling views:
  - Attribute views
  - Analytical views
  - Calculation views
- Read or write data to SAP Datasphere when you use a secure connection through the Secure Socket Layer (SSL) protocol.
  For more information about the SAP Datasphere service, see SAP Datasphere in the SAP documentation.

You can use SAP HANA objects as sources, targets, or lookups in mappings and mapping tasks. You can switch mappings to advanced mode to include transformations and functions that enable advanced functionality.

SAP HANA Connector assets

Create assets in Data Integration to integrate data using SAP HANA Connector.

When you use SAP HANA Connector, you can include the following Data Integration assets:

- Data transfer task
- Mapping
- Mapping task

For more information about configuring assets and transformations, see *Mappings*, *Transformations*, and *Tasks* in the Data Integration documentation.

CHAPTER 2

SAP HANA connections

Create an SAP HANA connection to connect to SAP HANA and SAP Datasphere. You can use an SAP HANA connection in mappings and mapping tasks to read from or write to SAP HANA and SAP Datasphere.

Prerequisites

Before you create an SAP HANA connection to read from or write to SAP HANA databases, the SAP HANA administrator needs to perform certain prerequisite tasks to configure the Secure Agent machine and SAP system.

1. Install the 64-bit SAP HANA ODBC driver.
2. Create an ODBC data source based on your operating system:
   - On Windows, use the ODBC data source administrator to create and configure a data source for the SAP HANA ODBC driver.
   - On Linux, add the entries to the `odbcinst.ini` file in the system `$ODBCHOME` directory.
3. Download and configure the `ngdbc.jar` file.

For more information about how to connect to SAP HANA databases in cloud, see Connecting to the SAP HANA database in SAP HANA Cloud in the SAP documentation.

After the administrator completes the configurations, you can set up and use an SAP HANA connection in mappings and mapping tasks.

Creating an SAP HANA ODBC data source on Windows

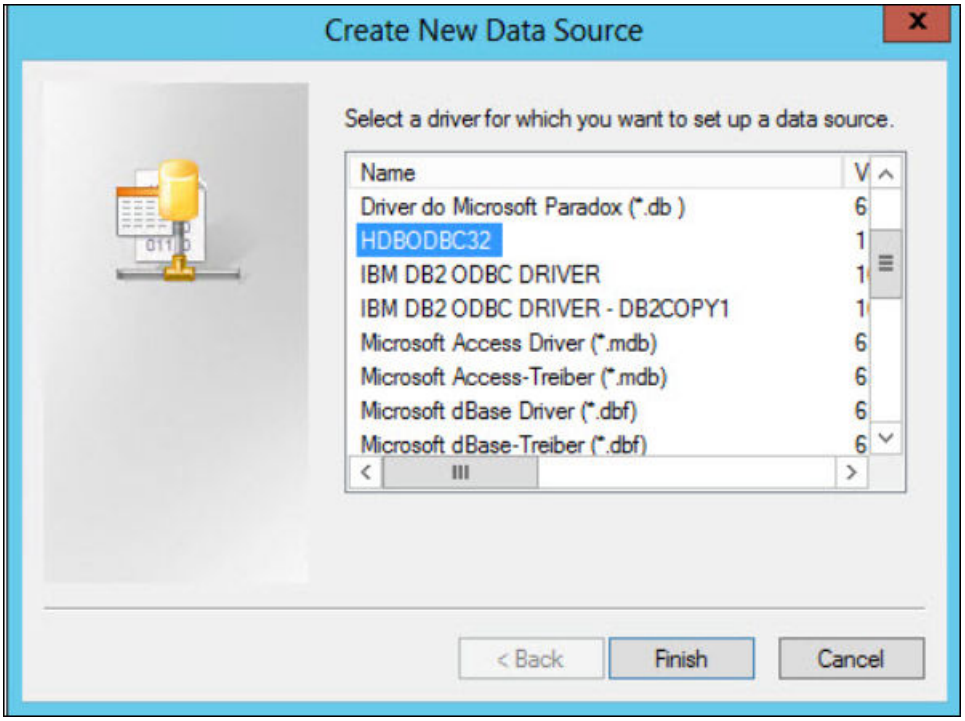
After you install the SAP HANA ODBC driver on Windows, use the ODBC data source administrator to create an SAP HANA ODBC data source.

1. Select **Start > Administrative Tools** > **Data Source (ODBC)**.

   The ODBC Data Source Administrator dialog box appears.
2. Click the **System DSN** tab, and then click **Add**.

   The Create New Data Source dialog box appears.

3. Select the HANA ODBC driver HDBODBC for 64-bit Windows or driver HDBCODBC32 for 32-bit Windows to set up the SAP HANA data source.

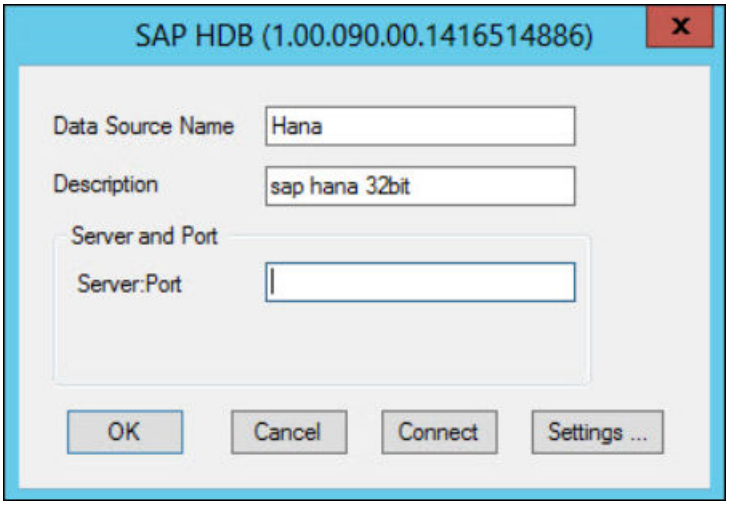


4. Click **Finish**.

   The SAP HDB dialog box opens.

5. Enter the data source name, description, server and port fields required to access SAP HANA.

   Additionally, click **Connect** to verify to test if the ODBC connection successfully connects to the SAP HANA server.



6. Click **OK** to complete the driver configuration.

Configure the ODBC driver on Linux

After you install the SAP HANA ODBC driver on Linux, configure the SAP HANA ODBC data source. The operating system user that runs the Informatica processes must have read and execute permissions on the SAP HANA ODBC driver file.

1. Set the following environment variables.
   ODBCHOME

   Set the variable to the ODBC installation directory.

   For example, `setenv ODBCHOME /export/home/Informatica Cloud Secure Agent/drivers/misc/latest/bin`

   ODBCINST

   Set the variable to the directory that contains the odbcinst.ini file.

   For example, `setenv ODBCINST /export/home/Informatica Cloud Secure Agent/odbcinst.ini`

   LD_LIBRARY_PATH, LIBPATH, or SHLIB_PATH

   Set the shared library environment variable to the directory where the SAP HANA ODBC driver is installed.

   For example, `setenv LD_LIBRARY_PATH ".:${PM_HOME}:${JAVA_HOME}/lib:${ORACLE_HOME}/lib:/usr/sap/hdbclient:"`
2. Add the SAP HANA driver details to the `odbcinst.ini` file located in the Secure Agent installation directory.
   For example, append the following SAP HANA driver details in the `odbcinst.ini` file to connect to the SAP HANA database:

   ```
   [HDBODBC]
   Driver=/usr/sap/hdbclient/libodbcHDB.so
   Description=HANA Driver
   Setup=/usr/sap/hdbclient/libodbcHDB.so
   CPTimeout=0
   ```

   Note: In the example, the following path is the location for the driver installation: `/usr/sap/hdbclient/`

Download and configure the SAP HANA libraries

SAP HANA Connector uses JDBC to import the metadata. Hence, to read data from the SAP HANA database, download the `ngdbc.jar` file and configure it on the Secure Agent machine. Contact SAP Customer Support if you encounter any issues with downloading the file.

1. Go to the SAP Service Marketplace: http://service.sap.com/connectors

   Note: You need SAP credentials to access the Service Marketplace.
2. Download the `ngdbc.jar` file on the Linux or Windows machine where the Secure Agent runs.

   Verify that you download the most recent version of the file.
3. Copy the `ngdbc.jar` file to the following directory:

   `C:\Program Files\Informatica Cloud Secure Agent\apps\Data_Integration_Server\ext\deploy_to_main\bin\rdtm-extra\HANA`

   Create the `deploy_to_main\bin\rdtm-extra\HANA` directory if it does not already exist.
4. Restart the Secure Agent.

Connect to SAP HANA

Let's configure the SAP HANA connection properties to connect to SAP HANA and SAP Datasphere.

Before you begin

Before you configure a connection, download and configure the SAP HANA libraries, and install and configure the 64-bit SAP HANA ODBC driver and data source on the Secure Agent machine.

Check out "Prerequisites" on page 8 to learn more about these tasks.

Connection details

The following table describes the basic connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. When you enable the secret vault in the connection, you can select which credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run tasks. Select a Secure Agent, serverless, or elastic runtime environment. For more information about how to configure a serverless environment, see "Configure the serverless runtime environment" on page 12. For more information about how to configure an elastic runtime environment, see "Configure the elastic runtime environment" on page 13.
Host	SAP HANA server host name.
Port	SAP HANA server port number.
Database Name	Name of the SAP HANA database.
Current Schema	SAP HANA database schema name. Specify **_SYS_BIC** when you use SAP HANA database modelling views.

Property	Description
Code Page	The code page of the database server defined in the connection. Select a code page from the following list: - MS Windows Latin1 - UTF-8 - Shift-JIS - ISO 8859-15 Latin 9 (Western European) - ISO 8859-2 Eastern European - ISO 8859-3 Southeast European - ISO 8859-5 Cyrillic - ISO 8859-9 Latin 5 (Turkish) - IBM EBCDIC International Latin-1
Username	User name of the SAP HANA account.
Password	Password of the SAP HANA account. The password can contain alphanumeric characters and the following special characters: ~ ` ! @ # $ % ^ & * ( ) _ - + = [ ] | : ; ' < , > . ? /

Advanced settings

The following table describes the advanced connection properties:

Property	Description
Metadata Advanced Connection Properties	The optional properties for the JDBC driver to fetch the metadata. If you specify more than one property, separate each key-value pair with a semicolon. For example, you can set the connection timeout for the JDBC driver when you connect to SAP HANA.
Run-time Advanced Connection Properties	The optional properties for the ODBC driver to run mappings. If you specify more than one property, separate each key-value pair with a semicolon. For example, `charset=sjis;readtimeout=180`

Configure the serverless runtime environment

You can use the serverless runtime environment to connect to the SAP system when you configure an SAP HANA connection in Data Integration.

Before you configure a SAP HANA connection using the serverless runtime environment, you need to perform certain prerequisites.

1. Create the following structure for the serverless agent configuration on AWS or Azure:
   `<Supplementary file location>/serverless_agent_config`
2. Add the libraries in the Amazon S3 bucket or Azure container in the following directory in your AWS or Azure account: `<Supplementary file location>/serverless_agent_config/hana`

3. Copy the following code snippet to a text editor:

```
version: 1
agent:
  dataIntegrationServer:
    autoDeploy:
      sap:
        hanas:
          - fileCopy:
              sourcePath: sap/jco/ngdbc.jar
      odbcInst:
        drivers:
          - fileCopy:
              sourcePath: ODBC/libodbcHDB.so
        dsns:
          - name: "HDBODBC"
            entries:
              - key: Driver
                value: libodbcHDB.so
              - key: Description
                value: "HANA Driver"
              - key: CPTimeout
                value: 0
```

   where the source path is the directory path of the library files on AWS or Azure.

4. Ensure that the syntax and indentations are valid, and then save the file as `serverlessUserAgentConfig.yml` in the following AWS or Azure directory:
   `<Supplementary file location>/serverless_agent_config`

   When the .yml file runs, the libraries are copied from the AWS or Azure directory to the serverless agent directory.

For more information about how to configure and use the serverless runtime environment, see *Serverless Runtime Environments* in the Administrator help.

Configure the elastic runtime environment

You can configure custom binary files in an elastic runtime environment to ensure that the runtime environment can access and execute these files during mapping runs.

Before you configure custom binary files, make sure that you deploy the elastic runtime environment on AWS to create a Kubernetes cluster that can process data within Informatica® Intelligent Data Management Cloud™ (IDMC).

For more information about deploying an elastic runtime environment, see Deploy an elastic runtime environment.

After you deploy the elastic runtime environment, place the custom binary files on the mounted disk and copy the exact path of the binary files to use in the POST request.

Perform the following steps within IDMC for the elastic runtime environment to manage the custom binary files:

1. Log in to your organization and obtain the session ID and runtime environment ID.
2. Make a POST call to the REST API, passing the session ID, runtime environment ID, and the binary file path copied earlier from the mounted disk.
   For more about information about making the POST calls, see Supplementary files in the REST API Reference guide.

The following is an example of the POST request:

```
POST <base URL>/api/v3/RuntimeEnvironment/<runtime environment ID>/ElasticConfig
Content-Type: application/json
Accept: application/json
INFA-SESSION-ID: <SessionId>
{
  "rteId": "rte-12345",
  "specification": {
    "agent": {
      "dataIntegrationServer": {
        "sap": {
          "jcos": [{"sourcePath": "/<path to binaries>/jco1"}],
          "nwrfcs": [{"sourcePath": "/<path to binaries>/nwrfc1"}],
          "hanas": [{"sourcePath": "/<path to binaries>/hana1"}]
        }
```

The POST call triggers a restart of the Data Integration Server.

3. Verify that the elastic runtime environment is up and running by checking the status of the Data Integration Server in Administrator.
4. Test a connection or run a mapping to confirm that the elastic runtime environment can access and use the custom binary files.

For more information about how to configure and use the elastic runtime environment, see *Elastic Runtime Environments* in the Administrator help.

Configure the Secure Socket Layer protocol

You can use the SAP HANA connection with the Secure Socket Layer (SSL) protocol to securely read from or write to SAP HANA.

You can also use this secure connection when you read from or write to SAP Datasphere.

For more information about the steps to configure an SAP HANA SSL connection, see Configure a secure SAP HANA connection Informatica How-To Library article.

CHAPTER 3

Mappings and mapping tasks with SAP HANA Connector

Use the Data Integration Mapping Designer to create a mapping. When you create a mapping, you configure a source or target to represent an SAP HANA object.

In advanced mode, the Mapping Designer updates the mapping canvas to include transformations and functions that enable advanced functionality.

Describe the flow of data from source and target along with the required transformations before the agent writes data to the target. When you create a mapping task, select the mapping that you want to use. Use the Mapping Task wizard to create a mapping task. Validate and run the mapping to read data from sources and write to a target. The mapping task processes data based on the data flow logic you define in the mapping.

SAP HANA sources in mappings

To read data from an SAP HANA database, configure an SAP HANA object as a source in a mapping.

Specify the name and description of the SAP HANA source. Configure the source, query options, and advanced properties for the source object.

You can parameterize the SAP HANA source connection at run time with values specified in a parameter file.

The following table describes the SAP HANA source properties that you can configure in a Source transformation:

Property	Description
Connection	Name of the source connection. You can select an existing connection, create a new connection, or define parameter values for the source connection property. To overwrite the source connection properties at run time, select the **Allow parameter to be overridden at run time** option.
Object Type	Type of the source object. Select one of the following object types from the list: - Single Object - Multiple Objects - Query - Parameter Default is Single Object.
Object	The source object for the task. Select the source object for a single source.
Filter	Filters record based on the filter condition in a read operation. A simple filter includes a field name, operator, and value. Use an advanced filter to define a more complex filter condition, which can include multiple conditions using the AND or OR logical operators. Use single quotes to specify a filter value.
Select distinct rows only	Select this option to extract only distinct rows.

The following table describes the SAP HANA source advanced properties that you can configure in a Source transformation:

Property	Description
Pre SQL	Pre-SQL command to run before reading data from the source.
Post SQL	Post-SQL command to run after reading data from the source.
Output is Deterministic	When you configure this property, the Secure Agent does not stage source data for recovery if transformations in the pipeline always produce repeatable data.
Output is repeatable	When the output is deterministic and the output is repeatable, the Secure Agent does not stage the source data for recovery.
SQL Override	The SQL statement that overrides the default query generated by the specified source object used to read data from the SAP HANA source.
Schema Name	Overrides the schema name specified in the connection.

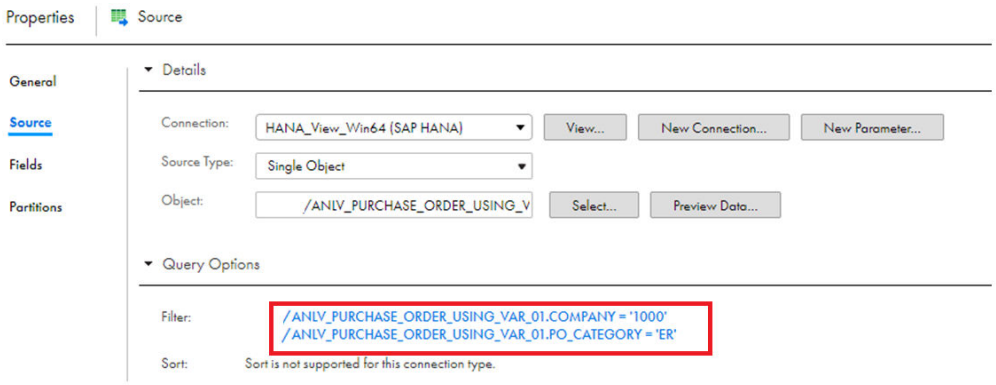
Property	Description
Table Name	Overrides the table name of the imported SAP HANA source table.
Tracing level	Sets the amount of detail that appears in the log file. Select one of the following options from the list: - Terse - Normal - Verbose Initialization - Verbose Data Default is Normal.

Input types for HANA modelling views

You can use the following input types when you read data from HANA modelling views:

- Variables. A variable is a filter. A variable is associated with a field, for example, company name, of the view.
  Provide the variable value in the field that you specified in SAP.

  The following image shows the variable `VAR_01` associated with `COMPANY` and `PO_CATEGORY` fields:



- Parameters. An input parameter is a placeholder to the query. You can use parameters to insert values for analytical views and calculation views. The data type for an input parameter in Data Integration is `char`. Specify the data for input parameters in the format that you defined in SAP.
  When you create a mapping and use input parameters, do not connect the input parameters in the source to the output fields. Otherwise, the mapping fails. For more information, see "Input parameter types" on page 17.

Input parameter types

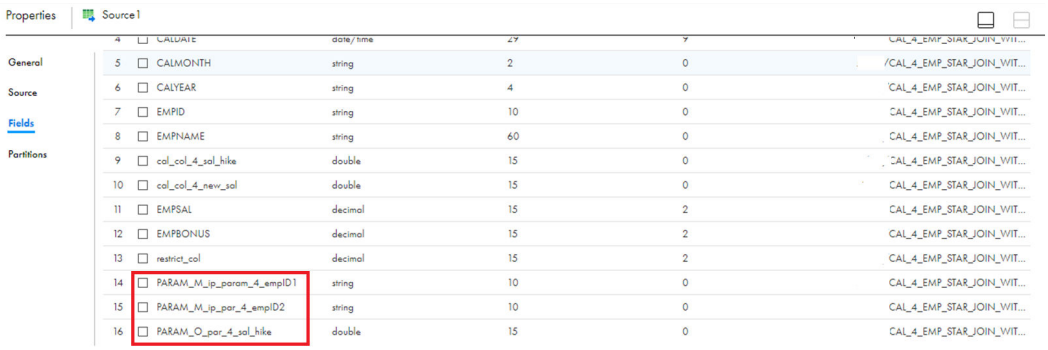
When you select a HANA modelling view, Informatica adds a prefix to the parameter name. The prefix is used to indicate the parameter type.

You can use the following types of parameters:

- Mandatory Parameter. A parameter for which you are required to specify a value. For example, in the field `PARAM_M_ip_param_4_empID1`, `PARAM_M` is the prefix for a mandatory parameter that Informatica adds. `ip_param_4_empID1` is the parameter name that is a part of HANA modelling views.

- Optional Parameter. A parameter for which you do not need to provide a value. For example, in the field `PARAM_O_par_4_sal_hike`, `PARAM_O` is the prefix for an optional parameter that Informatica adds. `par_4_sal_hike` is the parameter name that is a part of HANA modelling views.

The following image shows the mandatory parameter `PARAM_M_ip_param_4_empID1` and the optional parameter `PARAM_O_par_4_sal_hike`:



SAP HANA targets in mappings

To write data to an SAP HANA database, configure an SAP HANA object as a target in a mapping.

Specify the name and description of the SAP HANA target. Configure the target and advanced properties for the target object.

You can parameterize the SAP HANA target connection at run time with values specified in a parameter file.

The following table describes the SAP HANA target properties that you can configure in a Target transformation:

Property	Description
Connection	Name of the target connection. You can select an existing connection, create a new connection, or define parameter values for the target connection property. To overwrite the target connection properties at run time, select the **Allow parameter to be overridden at run time** option.
Object Type	Type of the target object. Select one of the following object types from the list: - Single Object - Parameter **Note:** Multiple objects target type is not applicable. Default is Single Object.
Object	The target object for the task. You can select an existing target object or create an SAP HANA target at run time with an object name that you specify.

Property	Description
Operation	The target operation. You can choose from the following operations: - Insert - Update - Upsert - Delete - Data Driven Default is Insert.
Truncate target	The Secure Agent truncates the target before writing the data.

The following table describes the SAP HANA target advanced properties that you can configure in a Target transformation:

Property	Description
Pre SQL	The SQL statement that you want to run before reading data from the source.
Post SQL	The SQL statement that you want to run after writing data into the target.
Update Override	The SQL statement that overrides the default query generated by the specified target object used to write data to the SAP HANA target.
Schema Name	Overrides the schema name specified in the connection.
Table Name	Overrides the table name of the imported SAP HANA target table.
Success File Directory	Not applicable.
Error File Directory	Not applicable.
Forward Rejected Rows	Determines whether the transformation passes rejected rows to the next transformation or drops rejected rows. By default, the agent forwards rejected rows to the next transformation.

SAP HANA lookups in mappings

When you configure field mappings, you can create a lookup to an SAP HANA object. Lookups are only applicable for SAP HANA database tables and views. You can't configure lookups for SAP HANA database modelling views.

You can parameterize the SAP HANA lookup connection at run time with values specified in a parameter file.

Note: You can't configure a Lookup transformation in a data transfer task.

The following table describes the SAP HANA lookup object properties that you can configure in a Lookup transformation:

Property	Description
Connection	Name of the lookup connection. You can select an existing connection, create a new connection, or define parameter values for the lookup connection property. If you want to overwrite the lookup connection properties at run time, select the **Allow parameter to be overridden at run time** option. When the task runs, the agent uses the parameters from the file that you specify in the task advanced session properties. You can switch between a non-parameterized and a parameterized SAP HANA connection. When you switch between the connections, the advanced property values are retained.
Object Type	Type of the source object. Select one of the following lookup object types from the list: - Single Object - Query - Parameter Default is Single Object. When the lookup source is large, you can use a custom query to reduce the number of columns to query.
Lookup Object	Name of the lookup object for the mapping.
Parameter	This property appears only if you select **Parameter** as the object type. A placeholder to dynamically assign values at run time for the lookup object without the requirement to edit the task. Select an existing parameter for the lookup object or click **New Parameter** to define a new parameter for the lookup object. If you want to overwrite the parameter at run time, select the **Allow parameter to be overridden at run time** option when you create a parameter. When the task runs, the agent uses the parameters from the file that you specify in the task advanced session properties.
Multiple Matches	Behavior when the lookup condition returns multiple matches. Select one of the following options from the list to determine the behavior: - Return first row - Return last row - Return any row - Return all rows - Report error Default is Return any row.
Query	This property appears only if you select **Query** as the object type. The query to look up data. Click **Define Query**, enter a valid custom query, and then click **OK**.

The following table describes the SAP HANA lookup advanced properties that you can configure in a Lookup transformation:

Property	Description
Lookup Source Filter[1]	Limits the number of lookups that the mapping performs on the cache of the lookup source table based on the value you specify in the filter condition. Maximum length of the filter condition is 32768 characters. This property is applicable when you select **Single Object** as the object type and enable lookup cache on the **Advanced** tab of the Lookup transformation. For more information about this property, see *Transformations* in the Data Integration documentation.
Lookup SQL Override	Overrides the default SQL query used to read data from the SAP HANA source.
Tracing Level[1]	Sets the amount of detail that appears in the log file. Select one of the following options from the list: - Terse - Normal - Verbose Initialization - Verbose Data Default is Normal.
[1]Doesn't apply to mappings in advanced mode.	

Rules and guidelines for SAP HANA mappings

Consider the following rules and guidelines when you configure an SAP HANA mapping:

- When you create a target using the **Create New at Runtime** option, you can't use records with the Time data types.
- You can't read or write data that contains Unicode characters in the SAP HANA table name or the column name.
- When you configure a mapping with the **Create New at Runtime** option and the source table type is a row or column, the Secure Agent creates a row-based table, irrespective of the source table type.
- When you create a target using the **Create New at Runtime** option, ensure that the length of any of the columns in the HANA target table must not exceed 5000 characters.
- You can't override the source and target object parameters in an SAP HANA mapping at run time with values specified in a parameter file.
- If you want to configure an advanced filter using both input parameters and field conditions to read data from an SAP HANA modelling view, the filter conditions must have the following query syntax:
  `"<Package name>/<Modelling view name>".<Input parameter name> = '<Input parameter value>' AND "<Package name>/<Modelling view name>"."<Field name>" = '<Field value>'`
- To run a mapping that reads data from or writes data to an SAP HANA table, ensure that the SAP HANA table name does not contain the double quote character at the beginning of the table name. Otherwise the mapping fails at run time.

- To filter data from an SAP HANA modelling view using a simple data filter that contains an integer type input parameter, enclose the input parameter value in single quotes in the filter expression.
- Don't configure nested in-out parameters in the query source type when you read from SAP HANA.
- When you configure a function in the **Query** or **Lookup SQL Override** property in the Lookup transformation, specify the alias name for the function with the lookup field as an argument.
  For example:

  ```
  SELECT first_name AS fname, last_name AS lname, age AS years_old FROM person;
  ```

  In this query, "fname", "lname", and "years_old" are the alias names.

Troubleshooting SAP HANA mappings configured with query objects

To troubleshoot issues with SAP HANA mappings that use query source objects, you need to configure the *EnhancedHANAQueryCapabilities* custom property for the Secure Agent.

This configuration can help you troubleshoot the following issues:

- When you import data of the Decimal data type from SAP HANA, the precision of a Decimal field increases by 1 compared to its actual precision.
  For example, if the actual precision of a Decimal field is 6, it displays 7 in the **Fields** tab of the **Properties** panel.
- When a schema name specified in the **Current Schema** field differs from the default schema name associated with the user name in an SAP HANA connection and you create a target at run time in a mapping, the Secure Agent ignores the schema name specified in the **Current Schema** field in an SAP HANA connection and creates the target table in the default schema for users.
  For example, when you specify the "CLOUD_USER" schema name in the **Current Schema** field and "HANA_USER" in the **Username** field in an SAP HANA connection, and create a target with the **Create New at Runtime** option, the Secure Agent creates the target table in the "HANA_USER" schema.
- When a schema name specified in the **Current Schema** field differs from the default schema name associated with the user name in an SAP HANA connection and the custom query does not include a schema name, the Secure Agent ignores the schema name specified in the **Current Schema** field in an SAP HANA connection and uses the default schema for users.
- Even though you specify an alias name for a column, the Secure Agent ignores the alias name for the column and displays the actual column name in the **Fields** tab.

To configure the *EnhancedHANAQueryCapabilities* custom property for the Secure Agent, perform the following steps:

1. Log in to Informatica® Intelligent Data Management Cloud™ (IDMC).
2. Select **Administrator** > **Runtime Environments**.
3. On the **Runtime Environments** page, select the Secure Agent.
4. Click **Edit**.
5. In the **Custom Configuration Details** section, select **Data Integration Server** as the service and **PMRDTM_CFG** as the type.
6. Enter *EnhancedHANAQueryCapabilities* in the **Name** field and set the value to *true*.

7. Click **Save**.

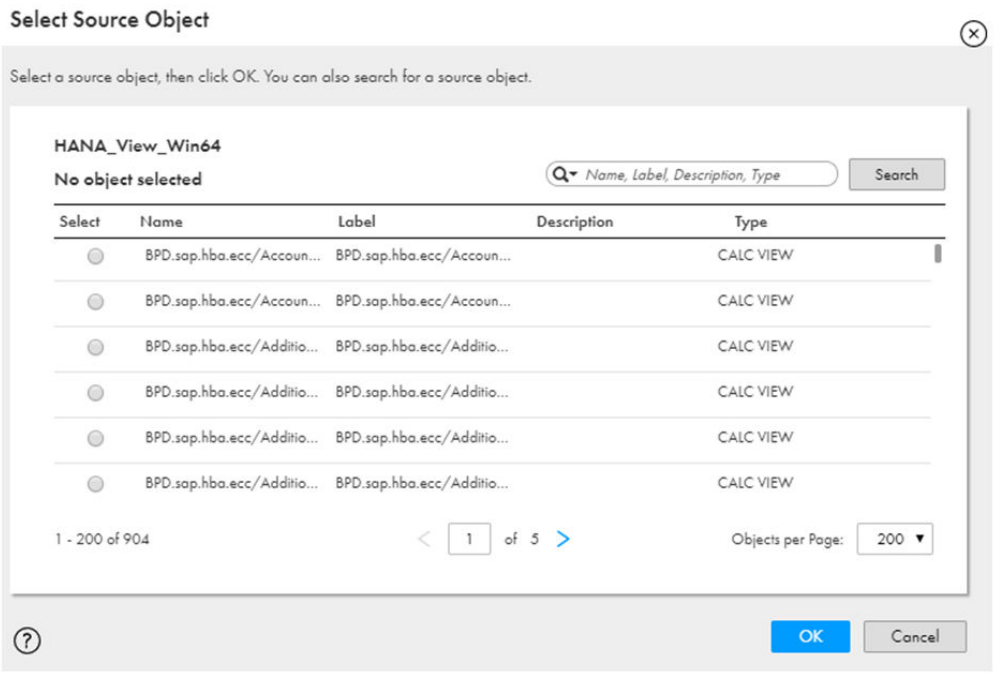
Creating an SAP HANA mapping with modelling views

You can use an SAP HANA object as a source in mappings and mapping tasks. Perform the following steps to create a mapping:

1. Click the Data Integration service.
2. Click **New** > **Mappings** > **Create**.
3. Click **Source** in the Design pane.
4. Click **Source** in the Properties pane.
5. Select a connection.
6. Select the **Source Type** as **Single Object**.
7. Click **Select** to select the object name.

   The **Select Source Object** page appears.
8. Select a source object from the list of objects.

   A list of objects is shown in the following image:



   The SAP HANA database modelling views for the source object that you created are displayed under the **Type** column. An analytical view is displayed as **OLAP VIEW**, an attribute view is displayed as **JOIN VIEW**, and a calculation view is displayed as **CALC VIEW**.
9. Click **OK**.
10. Add objects to the mapping as needed.
11. Save and run the mapping.

CHAPTER 4

SAP HANA SQL ELT optimization

You can configure SQL ELT optimization for an SAP HANA task to push the transformation logic to the source or target SAP HANA database. SQL ELT optimization enhances the mapping performance.

You can enable full or source SQL ELT optimization in an SAP HANA task that uses an SAP HANA connection in the mapping.

The amount of transformation logic that you can push to the database depends on the database, transformation logic, and task configuration. The task processes all transformation logic that it cannot push to the database.

You cannot configure SQL ELT optimization for mappings in advanced mode.

Configuring SQL ELT optimization for an SAP HANA mapping task

Perform the following steps to configure SQL ELT optimization for an SAP HANA mapping task:

1. Create an SAP HANA connection.
2. Create a mapping to read data from an SAP HANA source and write data to an SAP HANA target.
3. Create a mapping task.
   a. Select the configured mapping.
   b. In the **Runtime Options** page, in the SQL ELT optimization section, set the SQL ELT optimization value to **Full** or **Source**.
   c. Save the task and click **Finish**.

When you run the mapping task, the transformation logic is pushed to the SAP HANA database.

SQL ELT optimization supported operations, functions, operators, and transformations

The following table lists the supported operations:

Operation	SQL ELT optimization
Read	Yes
Write	Only the Insert operation is supported.

The following table summarizes the availability of functions that you can push to an SAP HANA database using full or source SQL ELT optimization.

Function	Function	Function
ABS()	INSTR()	SQRT()
AVG()	ISNULL()	STDDEV()
CHR()	LOG()	SUM()
CONCAT()	LOOKUP()	SYSDATE()
COS()	LOWER()	SYSTIMESTAMP()
COUNT()	MAX()	TAN()
DATE_COMPARE()	MIN()	TO_BIGINT
DATE_DIFF()	POWER()	TO_DECIMAL()
DATE_GET_PART()	ROUND()	TO_FLOAT()
DECODE()	SIN()	TO_INTEGER()
EXP()	SINH()	TRUNC()
IIF()	SOUNDEX()	UPPER()
-	-	VARIANCE()

The following table lists the operators that can be used in an SAP HANA database. Columns marked with an X indicate that the operator can be pushed to the SAP HANA database by using full **SQL ELT optimization**.

Operator
+
-

Operator
*
/
%
||
>
<
=
>=
<=
!=
AND
OR
NOT

The following table lists the transformations that you can push to SAP HANA:

Transformations
Aggregator
Expression
Filter
Joiner
Lookup
Router
Sorter
Union

CHAPTER 5

Data type reference

Data Integration uses the following data types in mappings and mapping tasks with SAP HANA:

SAP HANA native data types

SAP HANA data types appear in the Source transformation when you choose to edit metadata for the fields.

Transformation data types

Set of data types that appear in the transformations. They are internal data types based on ANSI SQL-92 generic data types, which the Secure Agent uses to move data across platforms. Transformation data types appear in all transformations in a mapping.

When Data Integration reads source data, it converts the native data types to the comparable transformation data types before transforming the data.

SAP HANA and transformation data types

SAP HANA data types map to transformation data types that the Secure Agent uses to move data across platforms.

The following table compares SAP HANA data types and transformation data types:

SAP HANA Datatype	Range	Transformation Datatype	Range
Alphanum	Precision 1 to 127	Nstring	1 to 104,857,600 characters
Bigint	-9,223,372,036,854,775,808 to 9,223,372,036,854,775,807	Bigint	-9,223,372,036,854,775,808 to 9,223,372,036,854,775,807 Precision 19, scale 0
Binary	Used to store bytes of binary data	Binary	1 to 104,857,600 bytes
Bintext	-	String	Precision 32000
Blob	Up to 2 GB	Binary	1 to 104,857,600 bytes
Boolean	True/False	Integer	Boolean (True/False) values, precision 10
Clob	Up to 2 GB	String	1 to 104,857,600 characters

SAP HANA Datatype	Range	Transformation Datatype	Range
Date	Jan 1, 0001 A.D. to Dec 31, 9999 A.D. Precision 10, scale 0	Date/Time	Jan 1, 0001 A.D. to Dec 31, 9999 A.D. (precision to the nanosecond)
Decimal (precision, scale) or Dec (p, s)	Precision 1 to 34	Decimal	Precision 1 to 28, scale 0 to 28 For transformations that support precision up to 38 digits, the precision is 1 to 38 digits, and the scale is 0 to 38. For transformations that support precision up to 28 digits, the precision is 1 to 28 digits, and the scale is 0 to 28. If you specify the precision greater than the maximum number of digits, the Secure Agent converts decimal values to double in high precision mode.
Double	Specifies a single-precision 64-bit floating-point number	Double	Precision 15
Float	Precision 1 to 53	Double	Precision 15
Integer	-2,147,483,648 to 2,147,483,647	Integer	-2,147,483,648 to 2,147,483,647 Precision 10, scale 0
NClob	Up to 2 GB	String	1 to 104,857,600 characters
Nvarchar	Precision 1 to 5000	String	1 to 104,857,600 characters
Real	Specifies a single-precision 32-bit floating-point number	Double	Precision 7, scale 0
Seconddate	0001-01-01 00:00:01 to 9999-12-31 24:00:00	Date/Time	Jan 1, 0001 A.D. to Dec 31, 9999 A.D. (precision to the nanosecond)
Shorttext	Specifies a variable-length character string, which supports text search and string search features	String	1 to 104,857,600 characters
Smalldecimal	Precision 1 to 16	Decimal	Precision 1 to 28, scale 0 to 28
Smallint	-32,768 to 32,767	Integer	Precision 5, scale 0
Text	Specifies a variable-length character string, which supports text search features	String	1 to 104,857,600 characters
Time	24-hour time period	Date/Time	Jan 1, 0001 A.D. to Dec 31, 9999 A.D. (precision to the nanosecond)

SAP HANA Datatype	Range	Transformation Datatype	Range
Timestamp	0001-01-01 00:00:00.0000000 to 9999-12-31 23:59:59.9999999	Date/Time	Jan 1, 0001 A.D. to Dec 31, 9999 A.D. (precision to the nanosecond)
Tinyint	0 to 255	Integer	Precision 5, scale 0
Varchar	Precision 1 to 5000	String	1 to 104,857,600 characters
Varbinary	1 to 5000 bytes	Binary	1 to 104,857,600 bytes

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